AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2013

Chemistry Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are SIX questions. Answer ALL questions. Questions 5 & 6 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 7 Marks)

a. $C_2H_4 + O_2 \Leftrightarrow CO_2 + H_2O$ (2 Marks)

i. Balance the given equation using the simplest possible whole numbers.

_____

_____

ii. How many moles of oxygen are required to react with 2.5 moles of ethene (C_2H_4)?

[Hint: Use balanced chemical equation for the calculation]

_____

_____

b. Draw the structures of any TWO isotopes of carbon specifying the number of protons, neutrons and electrons of each. (2 Marks)

Isotopes of Carbon	

c. Describe why sodium chloride has high melting and boiling points. (3 Marks)

Q.2. (Total 5 Marks)

a. Describe why electronegativity of elements increases from left to right in a period and decreases down a group in the periodic table. (4 Marks)

b. If an element belongs to group 17 and period 3 of the periodic table, what will be its electronic configuration? (1 Mark)

Q.3. (Total 9 Marks)

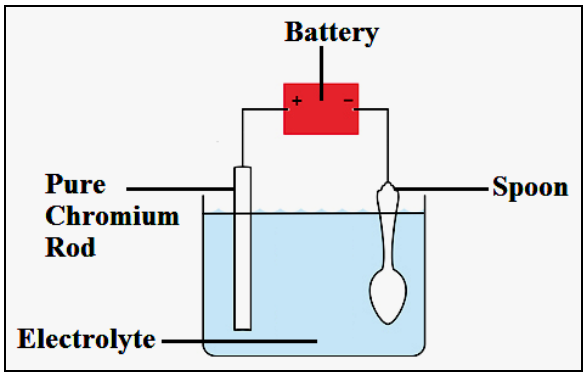
a. Kerosene oil and honey are both liquids. When added to water why does kerosene oil float over water while honey settles down? (2 Marks)

b. A student added potassium nitrate (KNO_3) in water and during dissolution of the salt he/she observed that the test tube became cold.

i. Describe why the test tube became cold. (2 Marks)

ii. What will happen to the solubility of this salt on increasing the temperature? (1 Mark)

c. The diagram below shows the electroplating of chromium.



i. Which of the two electrodes connected to the battery acts as a cathode? (1 Mark)

_____

_____

ii. Name the electrolyte used in the given electrolytic cell. (1 Mark)

_____

_____

iii. Show the reactions at cathode and anode using balanced chemical equations. (2 Marks)

Reaction at anode: _____

_____

Reaction at cathode:_____

_____

Q.4. (Total 4 Marks)

Give reasons for the following statements about metals and non-metals.

i. Non-metals do not react with dilute acids.

_____

_____

ii. Metals are electropositive in nature.

_____

_____

iii. Potassium is always found in nature as cations with +1 oxidation state.

_____

_____

iv. Fluorine is the most non-metallic among elements.

_____

_____

Q.5. (Total 8 Marks)

EITHER

a. Write any SIX characteristics in a tabular form distinguishing a compound from a mixture, giving ONE example of each.

OR

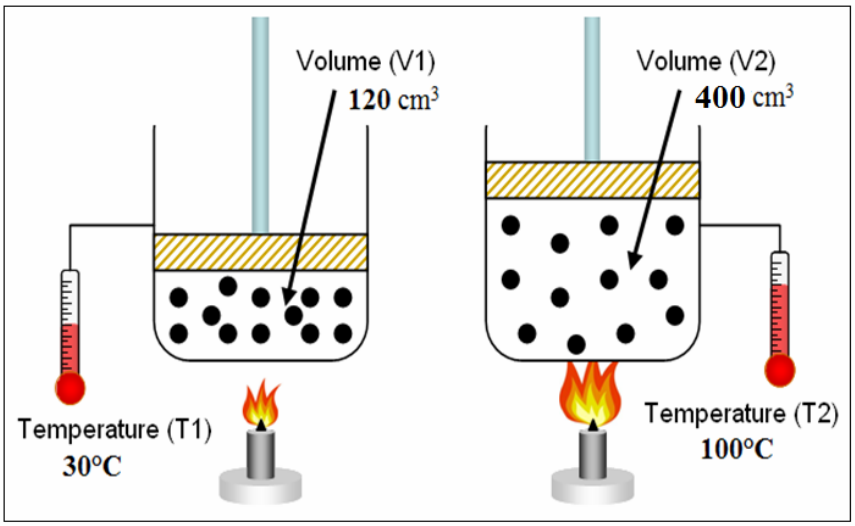
b. Describe the types of bonds and their formation in the following species. Also give ONE similarity between their bonds.

i. A water molecule (H_2O) ii. A hydronium ion (H_3O^+)

Q.6. (Total 7 Marks)

EITHER

a. The figure shows an experimental verification of a certain gas law with estimated values. Based on this experiment, describe the relevant gas law in detail showing mathematical and graphical representations of it.



OR

b. What is meant by the term ‘electroplating’? Describe in detail the electrolytic refining of impure copper showing reactions at anode and cathode with the help of chemical equations.

END OF PAPER

Please use this page for rough work